LSDN: A Lightweight Dual-Domain Attention Network With Adaptive Anchors for Ship Detection in Remote Sensing Scenarios

Shounan Wang, Chen Jia, *Member, IEEE*, Hui Liu, *Graduate Student Member, IEEE*, Yifan Yin, *Student Member, IEEE*, Xu Cheng, *Senior Member, IEEE*, and Shengyong Chen, *Senior Member, IEEE*

***Abstract*—Detecting small target ships (with pixel dimensions below 32 × 32) in remote sensing imagery is crucial for maintaining maritime traffic safety. However, achieving high accuracy with low computational cost remains challenging due to the dense scale distribution and the weak visual cues of small ship targets. To address these limitations, we propose a novel lightweight ship detection network (LSDN), which is capable of accurately localizing small ships in multiscale scenarios within remote sensing images, while maintaining a low parameter count and computational cost. Specifically, LSDN integrates three core components. First, the adaptive feature enhancement module (AFEM) employs a multilevel feature fusion strategy to enhance shallow feature representation, effectively preserving relevant information for small targets. Second, a lightweight backbone network, multiscale efficient strengthened attention module (MESA)-ShipDetection, incorporates a MESA to dynamically modulate feature importance across spatial and scale domains. Third, a dynamic anchor generation strategy constructs three sets of learnable anchor parameters based on target size distributions, thereby optimizing bounding box matching and localization accuracy. Experiments on the Airbus-Ship dataset demonstrate that LSDN achieves a detection accuracy of mean average precision (mAP) at 81.60% with only 3.65-M parameters, representing a 9.75% improvement over existing SOTA methods. In addition, ablation studies further validate the effectiveness of each module, with the AFEM increasing the recall rate of small targets by 7.83%, and the dynamic anchor strategy improving localization accuracy by 5.49%. This study offers a practical and efficient solution for real-time ship detection in resource-constrained scenarios.**

***Index Terms*—Feature enhancement, hybrid attention, lightweight model, remote sensing images, small ship detection.**

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Shounan Wang, Chen Jia, Hui Liu, Xu Cheng, and Shengyong Chen are with the School of Computer Science and Engineering, Tianjin University of Technology, Tianjin 300386, China (e-mail: jiachen@email.tjut.edu.cn; xu.cheng@ieee.org).

Yifan Yin is with the School of Computer Science, University of Technology Sydney, Sydney, NSW 9220, Australia.

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I. Introduction

MONITORING ships is of critical importance for ensuring maritime safety, managing marine resources, and protecting the environment [1]. Optical imagery from remote sensing offers crucial information for a variety of applications, including fisheries governance, maritime area organization, incident analysis, and environmental contamination management [2], [3]. The complex and dynamic maritime environment, however, often degrades image quality and alters object appearance [4], posing significant challenges for accurate ship detection.

Traditional detection methods primarily rely on background modeling and statistical inference, such as the CFAR algorithm [5], but their strong dependence on handcrafted features results in insufficient robustness and generalization capability in complex scenarios [6], [7]. Recently, data-driven deep learning methods have gradually superseded the traditional paradigm in ship detection, demonstrating immense potential across various remote sensing modalities. Taking synthetic aperture radar (SAR) as a prominent example, its all-weather imaging capability offers a unique advantage in ship detection [8]. However, it is challenged by inherent speckle noise and clutter interference. To address this, a multitude of methods have been proposed, including the use of large-kernel convolutions and feature fusion to mitigate multiscale imbalance [9], multitask learning and rotated bounding box detection to handle arbitrarily oriented targets [10], and even cutting-edge work that attempts to bypass the imaging process by detecting targets directly from raw echo data [11]. These explorations have not only advanced SAR detection performance but also provided new methodological insights for ship detection in optical remote sensing imagery.

In the domain of optical remote sensing, models based on convolutional neural networks (CNNs), such as VGG [12], Faster R-CNN [13], and the YOLO-series [14], have made remarkable progress in small object detection, particularly when integrated with structures like the feature pyramid network (FPN) [15] and single shot detectors (SSDs) [16] However, the inherent local receptive fields of CNNs limit their capability to model long-range dependencies, making it difficult to form global representations [17], [18], [19]. While the vision transformer (ViT) [20] addresses this issue

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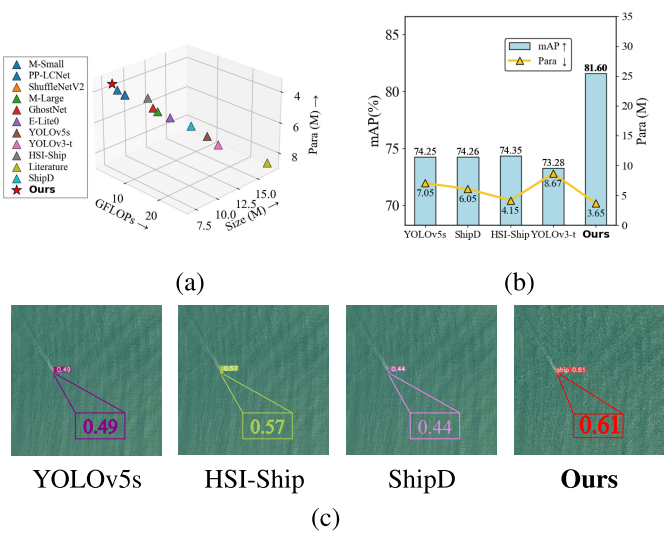


Fig. 1. Performance of LSDN on the Airbus-Ship dataset. (a) Comparison with SOTA methods. (b) Comparison of mAP and parameters. (c) Comparison of confidence scores under complex conditions.

by introducing a self-attention mechanism, its quadratic computational complexity and high memory demand constrain its application on resource-limited platforms [21]. Concurrently, existing network architectures commonly suffer from several issues, including large parameter scales, insufficient feature fusion, and limited shallow-layer representations [22], [23], [24], [25]. In the context of optical remote sensing for ship detection, these limitations manifest as three specific challenges: the tradeoff between model complexity and detection accuracy, inadequate feature representation for multiscale ship targets, and the severe loss of shallow-layer information crucial for small object detection [26], [27], [28]. Therefore, how to effectively model multiscale and small-object features while maintaining high detection accuracy and controlling model complexity remains a core challenge that needs to be addressed in this field.

To overcome the limitations, we introduce a lightweight ship detection network (LSDN) tailored for remote sensing applications. In this network, we have developed a lightweight network multiscale efficient strengthened attention module (MESA)-ShipDetection, to replace the complex backbone networks of traditional CNNs. To strengthen the model's feature representation across various scales, we introduced the hybrid attention mechanism MESA. This module refines spatial and multiscale information perception by integrating a spatial-multiscale dual-domain attention mechanism, effectively strengthening the model's feature representation capabilities for multiscale objects. In addition, to improve the extraction of shallow features, we designed the adaptive feature enhancement module (AFEM), which utilizes a multilevel feature fusion mechanism to enhance shallow feature extraction, thereby significantly improving the model's effectiveness in detecting small-scale ships. Finally, we construct three sets of learnable anchor parameters based on the actual target size, enabling dynamic optimization and matching of anchor box sizes and ratios. This further improves the model's capability to model multiscale objects. As shown in Fig. 1, our experiments utilizing a publicly available dataset for ship detection demonstrate that the proposed network achieves significant advantages in both detection accuracy and computational efficiency, fully supporting its utility in real-time operations and deployability on resource-constrained platforms. In summary, this article's main contributions included.

1) We propose an LSDN that focuses on resolving the challenges associated with small and scale-varying ships in optical remote sensing images, while maintaining computational efficiency for real-time and edge deployment.

2) We propose a lightweight backbone network named MESA-ShipDetection and proposed three task-aligned modules to enhance representation and localization: MESA that adaptively emphasizes spatial and scale-relevant features to improve the model's multiscale contextual representation capability, AFEM that enhances small ship object detection by reinforcing shallow features, and a dynamic anchor generation strategy that learns anchor shapes from data to further improve the localization accuracy of ships in multiscale, complex distribution scenarios.

3) We conduct a comprehensive evaluation on the Airbus-Ship dataset, where LSDN achieves superior performance in detecting small and multiscale ships with fewer than 4-M parameters, demonstrating its practicality for maritime monitoring under constrained computing conditions.

This article's subsequent sections are outlined below. Section II reviews contemporary research concerning ship detection. Section III describes the novel network proposed in this study. Section IV presents the experimental results. Section V concludes the research of this article.

II. Related Work

A. Lightweight Detection Models

Recently, substantial progress has occurred in designing and investigating lightweight neural network models. ShuffleNet [29], [30] employs pointwise group convolution in place of conventional pointwise operations and integrates a channel permutation mechanism. This dual method significantly curtails the model's parameter count while also overcoming the inherent restrictions of group convolution, all while preserving computational efficiency. Xception [31] optimizes Inceptionv3 by introducing depthwise separable convolution (DWConv), significantly enhancing model accuracy and processing speed, all while preserving its original parameter count. SqueezeNet [32] effectively reduces the number of model parameters by introducing a bottleneck structure. However, this design compromises the network's parallel computing capability while reducing parameters, resulting in relatively slower inference speed. MobileNet [33] not only optimizes VGGNet using DWConv but also develops the ReLU6 activation function specifically for lightweight networks, significantly enhancing the model's stability in low-precision computing environments. MobileNetv2 [34] introduces a linear bottleneck structure and inverted residual modules based on MobileNet [33], significantly improving both the precision and processing speed for image classification tasks. MobileNetv3 [35] leverages neural architecture search (NAS) to optimize the network parameters of MobileNet, further enhancing its performance.

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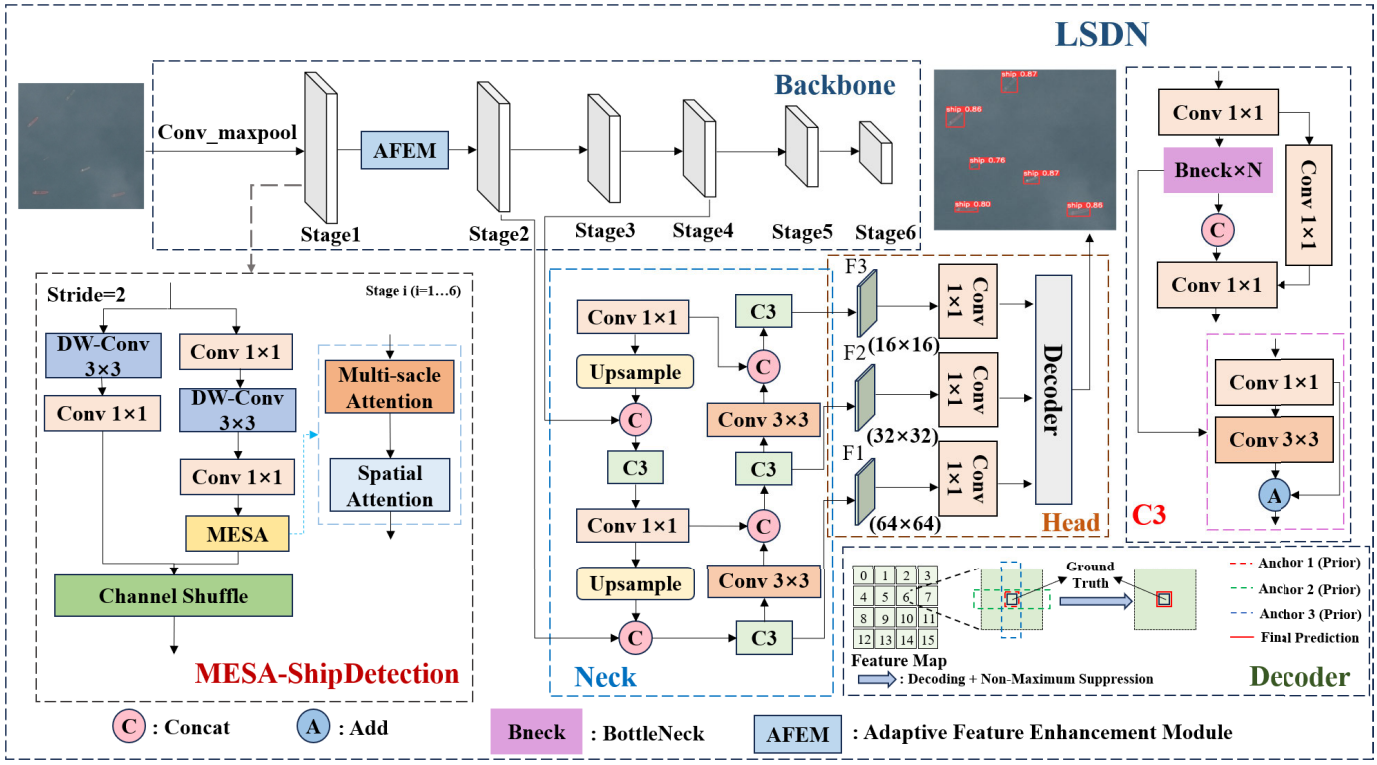


Fig. 2. Overview of the proposed LSDN architecture. The input image is first processed by the lightweight backbone MESA-ShipDetection, which integrates the AFEM module in its shallow stage to enhance small-target features and incorporates the MESA module to capture multiscale contextual information. The extracted features are then fused through the neck network to generate multiscale feature maps (F1, F2, and F3). These feature maps are subsequently fed into the YOLOv5-based detection head, which performs classification and bounding-box regression. Within the decoder, our dynamic anchor generation strategy provides adaptive anchors that optimize the final bounding-box predictions through decoding and nonmaximum suppression.

In the realm of ship identification, Wang et al. [36] developed a lightweight CNN architecture specifically designed for ship detection in infrared images. The model utilizes bottleneck DWConvs with residual connections to construct the backbone network, considerably lessening the computational load and enhancing its suitability for single-category object detection. Yin et al. [3] developed an improved architectural design leveraging CNNs, aimed at identifying ships. By introducing an improved convolutional module to replace S-Conv, the model effectively lowers the quantity of parameters whilst preserving detection accuracy. Zheng et al. [37] improved the YOLOv5 network by introducing the batch normalization (BN) scaling factor γ for compression, significantly enhancing detection accuracy and notably reducing computation time relative to alternative object detection algorithms.

B. Ship Detection Methods

Nowadays, CNNs have become the dominant method for detecting ships. Such as Changhua et al. [38] proposed a methodology for detecting small ships that leverages YOLOv3, which elevates the identification performance for small objects through adjustments to the design of the residual connections and the structure of the FPN. To overcome the multiscale limitation, Tian et al. [39] constructed an image augmentation unit leveraging a generative adversarial network (GAN) and additionally implemented a strategy to enlarge receptive fields. These innovations bolstered the model's capacity to capture salient features from target ships of varying scales.

In optical imagery, feature enhancement techniques are widely applied within ship detection tasks to address the issue of sparse feature information caused by the limited pixel count of ship targets and interference from complex backgrounds. Currently, common feature enhancement methods include multiscale feature fusion and attention mechanisms. The Transformer architecture, introduced by Vaswani et al. [40] signifies a landmark development within the realm of attention mechanisms. Its subsequent adaptation for visual tasks, which yielded the ViT [20], significantly enhances the model's expressive power and prediction accuracy through self-attention and multihead attention mechanisms. In this study, we introduce a novel LSDN for ships that integrates three core components: an AFEM module for enhancing shallow feature representation; a lightweight backbone network, MESA-ShipDetection, which incorporates the MESA module; and a dynamic anchor generation strategy that constructs anchor parameters based on target size distributions. Through experiments, we have shown that the proposed LSDN outperforms other methods in ship detection tasks. This method provides an efficient and practical solution for real-time ship detection in resource-constrained scenarios.

III. Methodology

A. Overview

Fig. 2 illustrates the architecture of our LSDN method, which comprises three core modules: a lightweight backbone network MESA-ShipDetection, which integrates the MESA

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module (detailed in Sections III-B and III-C); an AFEM that enhances shallow feature extraction capabilities to improve small object feature representation (detailed in Section III-D); and a dynamic anchor generation strategy that constructs anchor parameters based on the actual target size, achieving dynamic optimization and matching of anchor box sizes and ratios (detailed in Section III-E).

B. Lightweight Backbone Network MESA-ShipDetection

MESA-ShipDetection is a lightweight detection model designed to replace traditional CNN backbones. It significantly reduces computational requirements by replacing computationally intensive standard convolutions with DWConvs. Meanwhile, the channel shuffle operation efficiently facilitates interchannel information exchange at a very low computational cost. In addition, the model applies spatial downsampling with a stride of 2 in the early stages, further alleviating the computational burden of subsequent layers.

The model adopts a dual-branch architecture. In the left branch, when the stride is 2, downsampling and preliminary feature transformation are achieved through DWConv and 1×1 convolution. When the stride is 1, the left branch is skipped to simplify computation. The parameter count in this case can be expressed as

$$P = C_{\text{inp}} \times K^2 + 2 \times C_{\text{inp}} \times C_{\text{lb}} + 2 \times C_{\text{inp}} + 2 \times C_{\text{lb}} \tag{1}$$

where C_{inp} and C_{lb} are the respective channel counts for the input and left branch feature representations, and K is the size of the convolutional kernel.

In the right branch, the number of channels is first reduced through a 1×1 convolution to decrease computational complexity. Then, DWConv is used to extract more refined spatial features, followed by another 1×1 convolution for adjusting the channel dimensionality back to the target value. Finally, the MESA module is utilized to extract target features at different scales. The parameter count in this case can be expressed as

$$P = C_{\text{inp}} \times C_{\text{rb}} + C_{\text{rb}} \times \left(K^2 + 2 \times C_{\text{rb}}\right) + P_{\text{MESA}} \tag{2}$$

where C_{rb} is the channel count of the right branch feature representation, and P_{MESA} is the number of parameters in the MESA module.

Finally, the output feature maps of the left branch and right branch are concatenated along the channel dimension to integrate multibranch feature information. Subsequently, channel shuffling is applied to rearrange the channel order, promoting information interaction between different channels and avoiding information isolation, thereby enhancing model performance. This design, which combines a dual-branch architecture with channel shuffle, achieves an efficient balance between computational efficiency and feature representation capability, making it well-suited for lightweight object detection tasks.

C. Hybrid Attention Block MESA

To improve the model's ability to identify targets of varying scales, we propose a hybrid attention module named MESA. By integrating a spatial–channel dual-domain attention mechanism, it dynamically adjusts the importance of features across spatial and scale domains, enabling the model to adaptively focus on critical information in different scales and spatial regions, thereby improving detection accuracy for multiscale targets.

1) Multiscale Attention Block: Our proposed multiscale attention block (MAB) is designed to efficiently capture rich contextual information, with its architecture inspired by an in-depth analysis and fusion of existing advanced attention mechanisms. The classic coordinate attention (CA) successfully embeds precise positional information and captures long-range dependencies through 1-D feature encoding. However, its complete reliance on 1×1 convolutions results in a limited receptive field, making it deficient in modeling local multiscale context. On the other hand, the spatial attention branch of CBAM demonstrates the effectiveness of using larger kernels to capture local context, yet its sequential structure and shared-convolution design are not optimal. To overcome these limitations by synergizing their respective strengths (adopting the strengths and overcoming the weaknesses), we designed a parallel dual-branch structure. The first branch inherits the core idea of CA, employing 1-D average pooling (W_AvgPool and H_AvgPool) to capture direction-aware long-range dependencies that are crucial for target localization. To specifically compensate for this branch's limited receptive field, we designed a second parallel branch that directly utilizes a 3×3 convolution to efficiently extract local, multiscale features of the target, such as textures and fine-grained edge structures. Distinct from the static fusion methods of CA and CBAM, we introduce a data-driven, adaptive weighted fusion strategy. Through our designed adaptive weighting module (AS: AvgPool + Softmax) and matrix multiplication (Matmul), the long-range dependency features from the first branch and the local context features from the second are intelligently combined via a weighted sum. This design enables our module to autonomously balance the importance of global and local information based on the input content, ultimately generating a more powerful and discriminative feature representation, F'.

As shown in Fig. 3, the structure of the MAB differs from traditional sequential processing methods by adopting a parallel structure, effectively avoiding computational bottlenecks and improving model efficiency. First, the input feature map F is split into g groups, with each group having c//g channels, while maintaining the height and width unchanged.Next, adaptive average pooling is applied to the grouped feature maps along the width and height directions, yielding X_w and X_h, respectively. These are then concatenated along the height dimension to form H_w, followed by a 1×1 convolution operation to split them back into X_w and X_h. Subsequently, these two feature maps are processed through the Sigmoid activation function and element-wise multiplied with the original grouped feature maps to obtain the weighted feature maps.

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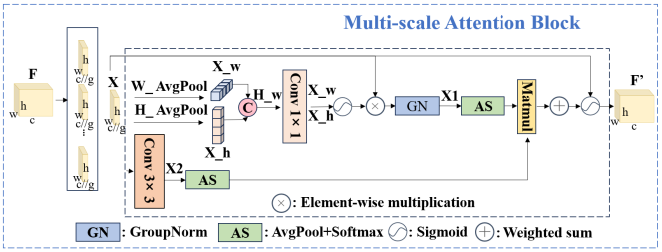


Fig. 3. Structure of MAB. Multiscale features are extracted via group convolution, while adaptive average pooling and 1 × 1 convolution capture spatial contextual information. Adaptive weight adjustment and matrix multiplication are then employed to fuse these multiscale features, incorporating an attention mechanism to enhance its attention to pertinent information.

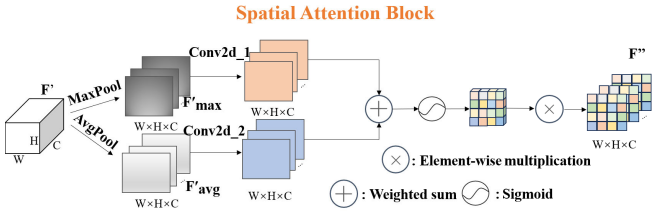


Fig. 4. Structure of SAB. The SAB generates an attention map by processing input feature map F' through parallel max pooling and average pooling branches, followed by convolutional layers and a sigmoid function. This attention map is then applied element-wise to F' to produce the spatially enhanced output F''.

Following this, group normalization is applied to the weighted feature maps to produce X_1. Meanwhile, the input feature map is processed through a 3×3 convolution to obtain X_2. Both X_1 and X_2 undergo adaptive average pooling, and their weights are computed using the Softmax activation function. These weights are then fused with their respective feature maps through matrix multiplication to generate weighted feature maps. Finally, these features undergo weighted summation and are then processed with the original grouped feature maps through a Sigmoid activation function to generate the weighted feature map F'.

2) Spatial Attention Block: Our proposed spatial attention block (SAB) is an improvement on the spatial attention mechanism in the classic attention mechanism, CBAM. We agree with CBAM's core insight that concurrently using average-pooling to capture global context and max-pooling to preserve salient features is crucial for generating high-quality spatial attention maps. However, we believe that CBAM's implementation has two optimizable limitations: first, it pools along the channel dimension, irreversibly compressing C channels of information into a single-channel representation, which may cause the loss of unique interchannel spatial relationships; second, it ultimately generates a 2-D spatial attention map that is shared by all channels, ignoring the characteristic that different channels may need to focus on different spatial regions. To solve these limitations, our SAB has made two key structural improvements. First, we completely preserve the channel dimension; the MaxPool and AvgPool in the module both operate on the spatial dimensions, ensuring that the information of the C channels is retained throughout the entire attention calculation process. Second, we introduce independent, unshared convolutional paths (Conv2d_1 and Conv2d_2) for these two pooling branches to learn more targeted feature transformations. This design can be seen as a generalization and enhancement of the CBAM idea: it no longer generates a generic 2-D spatial attention map for the entire feature map, but instead generates a dedicated 2-D spatial attention map for each channel, thereby enabling the learning of finer and more complex interactions between channels and space, and achieving superior performance.

As shown in Fig. 4, the module initially performs both maximum and average pooling upon the output activation tensor $F' \in R^{C \times H \times W}$ from the MAB in order to extract different spatial features. This results in F'_{max}, $F'_{avg} \in R^{C \times H' \times W'}$. The two outputs are then passed through independent 2-D convolutional layers to expand the receptive field. Afterward, they undergo fusion through an element-by-element addition. This fused representation is then passed through a Sigmoid nonlinearity. Finally, this outcome undergoes element-wise multiplication with the input features F', yielding the resultant F''. The final feature map F'' can be expressed as

$$F'' = \sigma\left(C2D\left(MP\left(F'\right)\right) \oplus C2D\left(AP\left(F'\right)\right)\right) \otimes F' \quad (3)$$

where σ represents the Sigmoid activation function, and MP and AP stand for the operations of maximum and average pooling, respectively.

D. Adaptive Feature Enhancement Module

To overcome the challenge of preserving adequate details from small targets, a consequence of the insufficient expressive power of shallow features in neural network models, we designed the AFEM. This module strengthens the shallow feature extraction capability through a multilevel feature fusion mechanism, effectively improving the detection performance of small targets. The structure of this module is illustrated in Fig. 5.

The module adopts a multibranch structure, where the input feature map F is processed through three main branches to extract multiscale and multitype features. The first branch is used to extract basic features, while the other two branches combine asymmetric convolutional kernels and dilated convolutions to expand the receptive field and capture contextual information. Simultaneously, the original input F undergoes a linear transformation through an independent 1×1 convolutional layer, forming a residual connection path to ensure the preservation of the original feature information. The features from the three branches are concatenated along the channel dimension and then fed into the AdaptiveFeatureFusion Module, where the multibranch features are adaptively weighted and fused. Subsequently, the fused features pass through a 1×1 convolutional layer to compress the channel dimension and are then combined with the linearly transformed residual branch features through weighted summation. Finally, the output feature map F' is obtained by applying the ReLU activation function. The AFEM module can ultimately be expressed as

$$AF\left(x\right) = FC\left(x\right) \otimes AG\left(FC\left(x\right)\right) \quad (4)$$

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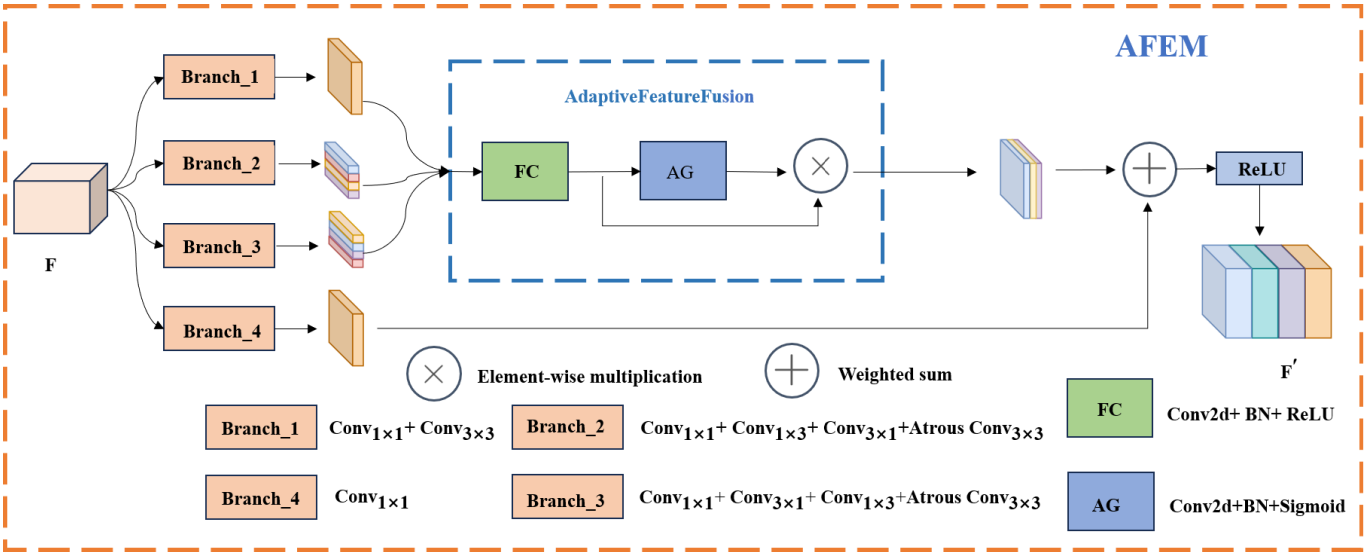


Fig. 5. Structure of AFEM. The input feature map F is processed through four distinct branches to extract features at varying scales. Subsequently, the outputs of these branches are fed into the adaptive feature fusion module, which adaptively weights and combines these features. Finally, the fused feature map is integrated with a residual connection and passed through a ReLU activation function to generate the enhanced feature map.

$$B_1 = AF\left(Conv_{3\times3}\left[Conv_{1\times1}(F)\right]\right) \tag{5}$$

$$B_2 = AF\left(DiConv_{3\times3}\left[Conv_{3\times1}\left[Conv_{1\times3}\left[Conv_{1\times1}(F)\right]\right]\right]\right) \tag{6}$$

$$B_3 = AF\left(DiConv_{3\times3}\left[Conv_{1\times3}\left[Conv_{3\times1}\left[Conv_{1\times1}(F)\right]\right]\right]\right) \tag{7}$$

$$F' = Cat(B_1, B_2, B_3) \oplus Conv_{1\times1}(F) \tag{8}$$

where FC denotes that the input values first undergo feature extraction and channel transformation through a 2-D convolutional layer (Conv2d), followed by BN to enhance generalization capability, and finally pass through a ReLU activation function to introduce nonlinearity. AG indicates that the input values first undergo feature extraction through a 2-D convolutional layer (Conv2d), followed by BN to enhance generalization capability, and finally pass through a Sigmoid activation function to generate attention weights.

E. Dynamic Anchor Generation Strategy

Traditional anchor generation methods typically employ fixed sizes and aspect ratios, which struggle to effectively address multiscale object detection problems, particularly in scenarios with significant scale variations among targets. Fixed anchors often fail to simultaneously match the characteristics of both large and small targets. To address this limitation, this article proposes a dynamic anchor generation strategy, which constructs three sets of learnable anchor parameters (with three anchors in each set) based on the actual dimensions of the targets, achieving dynamic optimization and matching of anchor box sizes and aspect ratios. Specifically, the K-means clustering algorithm is applied to analyze the target bounding boxes [ground truth boxes (GT Boxes)] in the training data, generating dynamic anchor parameters. Using the width and height of the target bounding boxes as feature vectors and adopting one-intersection over union (IoU) distance as the clustering metric, the generation of cluster centers is optimized. The dynamic anchor generation process can be expressed as

$$(w_k, h_k) = \arg\min_{(w,h)} \sum_{i=1}^{N} \min_{k} \left(1 - \mathrm{IoU}\left(g_i, (w_k, h_k)\right)\right) \tag{9}$$

where N represents the number of target bounding boxes in the training data, and $\mathrm{IoU}(g_i, (w_k, h_k))$ denotes the IoU between the ith target bounding box g_i and the kth cluster center (w_k, h_k), and k ranges from 1 to 9, corresponding to the three sets of dynamically generated anchor points.

To intuitively compare the differences in target size fitting between our approach and the original YOLOv5 anchors, we conducted a visual comparative analysis on the two datasets used in this article. First, as shown in Fig. 6, we qualitatively evaluated the anchor distributions using scatter plots. It can be clearly observed that in the Airbus-Ship training set [Fig. 6(a) and (b)], the distribution of the original anchors is sparse, with multiple large-scale anchors severely deviating from the core dense region of the GT Boxes. This leads to insufficient coverage capability for small targets. In stark contrast, our dynamic anchors, generated via the K-means algorithm, are highly focused on the central cluster of the data points, demonstrating a superior data fitting capability. In the LEVIR-Ship training set [Fig. 6(c) and (d)], only three of the nine original anchors are situated within the primary target region. The remainder are completely detached from the target cluster, failing to provide effective initial matches for the vast majority of instances. In contrast, our proposed anchor set demonstrates a superior data fitting capability. It is noteworthy that a few anchors in our set are also allocated to relatively sparse peripheral areas of the distribution. This is not an algorithmic flaw but rather a natural consequence of the K-means algorithm's objective to cover the few outlier targets with unique dimensions present in the dataset. This design ensures that our anchor set not only accurately represents the majority

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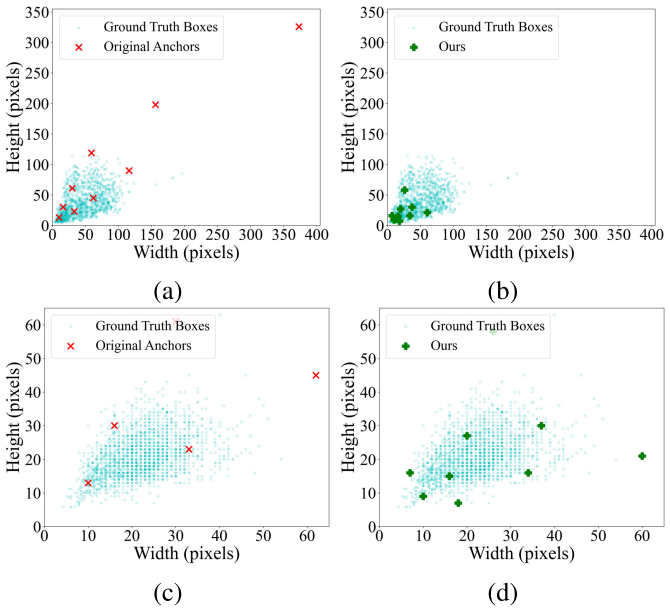


Fig. 6. Visualization of anchor distributions versus GT Box sizes. (a) and (b) Distribution comparison of the original anchors and the improved anchors against the GT Box sizes on the Airbus-Ship training set. (c) and (d) Same comparison on the LEVIR-Ship training set.

of common-sized targets but also provides effective proposals for the minority of special-sized ones, thereby enhancing the model's overall detection capability.

To quantitatively evaluate the generalization matching capability of the different anchor sets, we calculated the maximum possible IoU for each GT Box in the test set against both anchor groups and plotted their probability density distributions, as shown in Fig. 7. The analysis reveals that our dynamic anchor strategy exhibits a significant advantage: not only is its average max IoU substantially higher than that of the original anchors, but its entire distribution curve also shows a distinct rightward shift, with its peak occurring in a higher IoU range. These data clearly indicate that our dynamic anchor strategy possesses superior performance and generalization potential in matching GT Boxes compared to the original anchors.

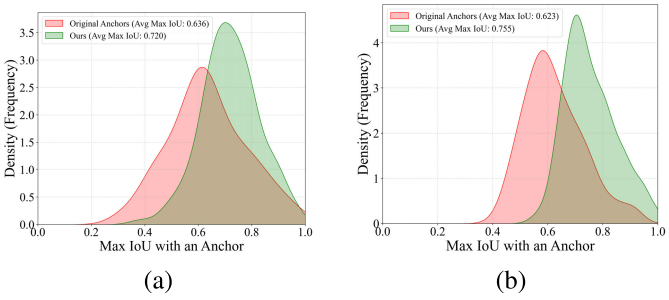


Fig. 7. Quantitative comparison of anchor matching quality on different datasets. (a) Probability density distributions of the best possible IoU for the original versus our proposed anchors on the Airbus-Ship test set. (b) Corresponding distributions on the LEVIR-Ship test set.

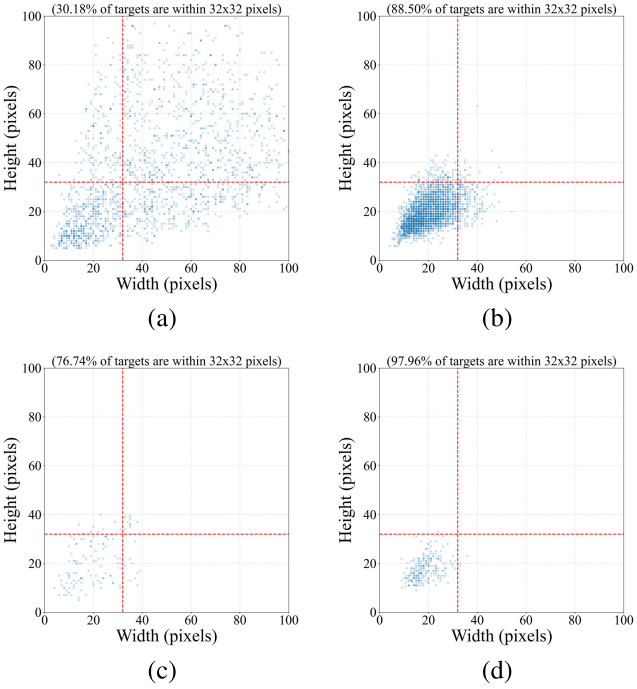


Fig. 8. Distribution of ship target sizes across different datasets and splits. The red dashed lines represent the 32 × 32 pixel threshold for small objects as defined in the COCO benchmark. (a) Size distribution for the entire Airbus-Ship dataset, while (c) distribution for its test set. (b) and (d) Distributions for the entire LEVIR-Ship dataset and its test set, respectively.

IV. Experiments

A. Datasets

1) Airbus-Ship[1]: The dataset comprises optical remote sensing imagery depicting ships, with each image dimensioned at 768 × 768 pixels, totaling 29 GB in size. For the ship target detection task in this study, we selected 1000 valid image samples containing small ships and divided them into training (700 images), validation (200 images), and test (100 images) sets at a ratio of 7:2:1.

2) LEVIR-Ship [41]: The dataset consists of 3896 satellite images (512 × 512 pixels) of small ships captured by China's GaoFen series satellites (GaoFen-1/6), featuring optical remote sensing capabilities with a spatial resolution of 16 m. After rigorous quality screening, 1900 high-quality samples were selected and subsequently allocated to respective training (70%), validation (20%), and test (10%) collections, with this division following a 7:2:1 proportion to ensure balanced data distribution.

To clearly define the scope of small targets in this study, we conducted a quantitative analysis of the target size distribution across the two experimental datasets. We adopted the widely used small object definition from the COCO [50] benchmark. Specifically, in this article, ships with a pixel size of less than 32 × 32 are defined as small ships. First, in the Airbus-Ship dataset, the overall target sizes exhibit a broad scale range, as shown in Fig. 8(a), with 30.18% of the samples strictly conforming to the small object definition. In its corresponding test set, the proportion of small targets increases substantially to 76.74%, as shown in Fig. 8(c). Second, in the LEVIR-Ship dataset, the targets are inherently concentrated in the

[1]https://www.kaggle.com/c/airbus-ship-detection

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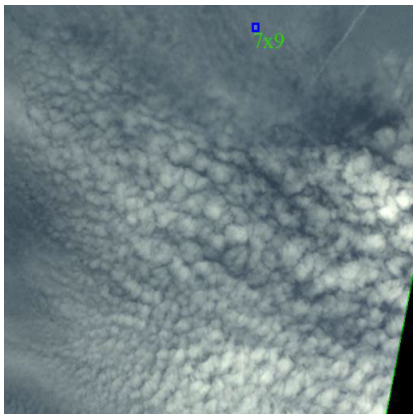


Fig. 9. Typical example of a small ship target in the LEVIR-Ship dataset. The target, measuring only 7×9 pixels.

small-size range. The small object proportion in the overall dataset reaches 88.50% [Fig. 8(b)], and this ratio further increases to an exceptionally high 97.96% in its test set [Fig. 8(d)]. This series of analyses robustly demonstrates that our experimental evaluation is rigorously focused on the core challenge of small object detection. However, it is crucial to emphasize that our definition of small targets in this article emphasizes the challenges of their pixel-level representation over their physical dimensions. These targets are typically difficult to recognize in remote sensing images due to their sparse pixel footprints and faint features. For example, as shown in Fig. 9, in the 16-m/pixel resolution LEVIR-Ship dataset, a target of just 7×9 pixels can correspond to a physical size of 112×144 m. This immense scale disparity underscores the essence of the detection task: the ability to extract discriminative features from severely constrained pixel information. This analysis provides the theoretical underpinning for our approach of treating small object detection as a pixel-level recognition problem.

B. Implementation Details

1) Experimental Settings: All empirical evaluations detailed herein were performed using a computational platform comprising an NVIDIA GeForce RTX 4090 GPU with 24 GB of memory, and the training realization relied on the PyTorch v1.8.0 framework alongside Python v3.8.0. Stochastic gradient descent (SGD) served as the optimizer, configured with a batch size of 4, where momentum and weight decay were set to 0.937 and $5e^{-4}$, respectively. The starting value for the learning rate was set to 0.01, and the model underwent training for 500 epochs. The final evaluation was performed using the model that demonstrated optimal performance on the validation set.

2) Comparison Methods: To validate the effectiveness of our method, we conducted comparative experiments with 15 SOTA methods. These include mainstream lightweight network architectures such as MobileNetV3-Small [35], PP-LCNet [42], ShuffleNetV2 [30], MobileNetV3-Large [35], GhostNet [43], EfficientNet-Lite0 [44]; lightweight detection models from the YOLO series like YOLOv5s, YOLOv3-tiny [45], and the newer YOLOv11s, YOLOv12s [46], and YOLOv13s [47]; the Transformer-based model OWRT-DETR [48]; as well as lightweight models specifically designed for the small ship detection task, such as HSI-ShipDetectionNet [28], Literature [49], and ShipDetectionNet [3].

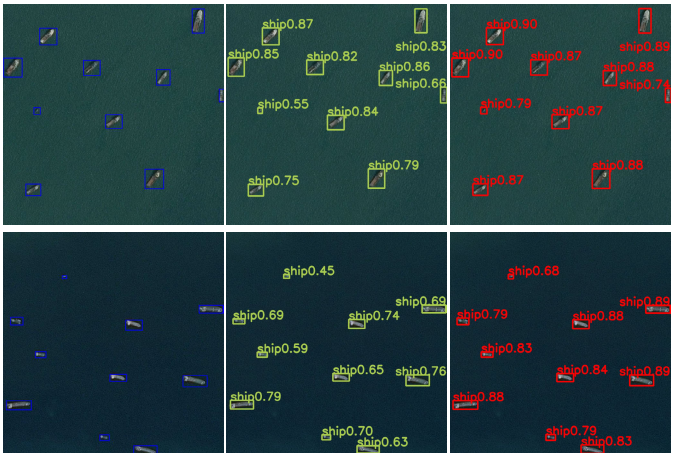


Fig. 10. Visualization comparison between our proposed method and HSI-ShipDetectionNet in dense scale distribution scenarios.

3) Evaluation Metrics: To comprehensively evaluate the model's effectiveness in detecting small ships, our evaluation metrics encompass multiple aspects, including the model's size, mean average precision (mAP), parameter count (Para), the calculated amount (GFLOPs), and the frames per second (FPS).

C. Comparison With SOTA Methods

To ensure all experiments were conducted under identical conditions, we standardized the images to a size of 512×512 pixels and adopted YOLOv5 (excluding the YOLO-series models and the OWRT-DETR model) as the unified detection framework for the aforementioned methods.

As shown in Table I, the proposed method exhibits significant advantages in both detection accuracy and computational efficiency. Compared with the current SOTA HSI-ShipDetectionNet, our method achieves a 12.05% reduction in parameters and a 75% decrease in GFLOPs, while simultaneously demonstrating remarkable improvements in detection accuracy. Notably, on the Airbus-Ship dataset, our method obtains a 9.75% increase in mAP. These advancements primarily stem from the design of the AFEM module's multi-level feature fusion mechanism, which substantially enhances the shallow network's feature extraction capability for small targets. Although YOLOv11s shows distinct advantages in FPS, its excessive parameter size and complex model architecture make it challenging to achieve efficient deployment in resource-limited edge environments.

To clearly demonstrate the effectiveness of our method, inference evaluations were carried out on selected images from the utilized datasets. As shown in Fig. 10, in such scenarios characterized by dense scale distribution with significantly varied scales, our model not only accurately detected all targets but also achieved higher confidence scores for every prediction. This demonstrates the effectiveness of our pro-

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TABLE I
PERFORMANCE COMPARISON WITH SOTA METHODS ACROSS TWO BENCHMARK DATASETS

Methods	Para(M) ↓	GFLOPs ↓	Size(MB) ↓	LEVIR-Ship mAP(%)	Airbus-Ship mAP(%)	FPS ↑
MobileNetV3-Small [35]	**3.54**	6.3	**7.2**	61.15	68.61	61
PP-LCNet [42]	3.74	8.1	7.4	60.68	70.87	72
ShuffleNetV2 [30]	3.78	7.7	7.5	59.90	68.86	59
MobileNetV3-Large [35]	5.20	10.3	10.2	71.46	70.52	51
GhostNet [43]	5.20	8.4	10.4	65.58	71.94	53
Efficient-Lite0 [44]	5.71	11.5	11.2	66.40	70.08	61
YOLOv5s	7.05	16.3	13.7	67.64	74.25	87
YOLOv3-tiny [45]	8.67	12.9	16.6	66.09	73.28	120
YOLOv11s	9.43	21.5	54.5	68.40	76.01	**157**
YOLOv12s [46]	9.30	19.6	52.8	68.12	76.53	93
YOLOv13s [47]	9.02	21.0	52.6	68.45	76.70	107
OWRT-DETR [48]	32.81	108.0	125.9	70.75	77.91	106
HSI-ShipDetectionNet [28]	4.15	10.0	9.2	72.37	74.35	53
Literaure [49]	8.65	26.1	17.0	67.15	70.47	76
ShipDetectionNet [3]	6.05	15.7	12.0	71.82	74.26	58
Ours	3.65	**2.5**	8.0	**72.39**	**81.60**	62

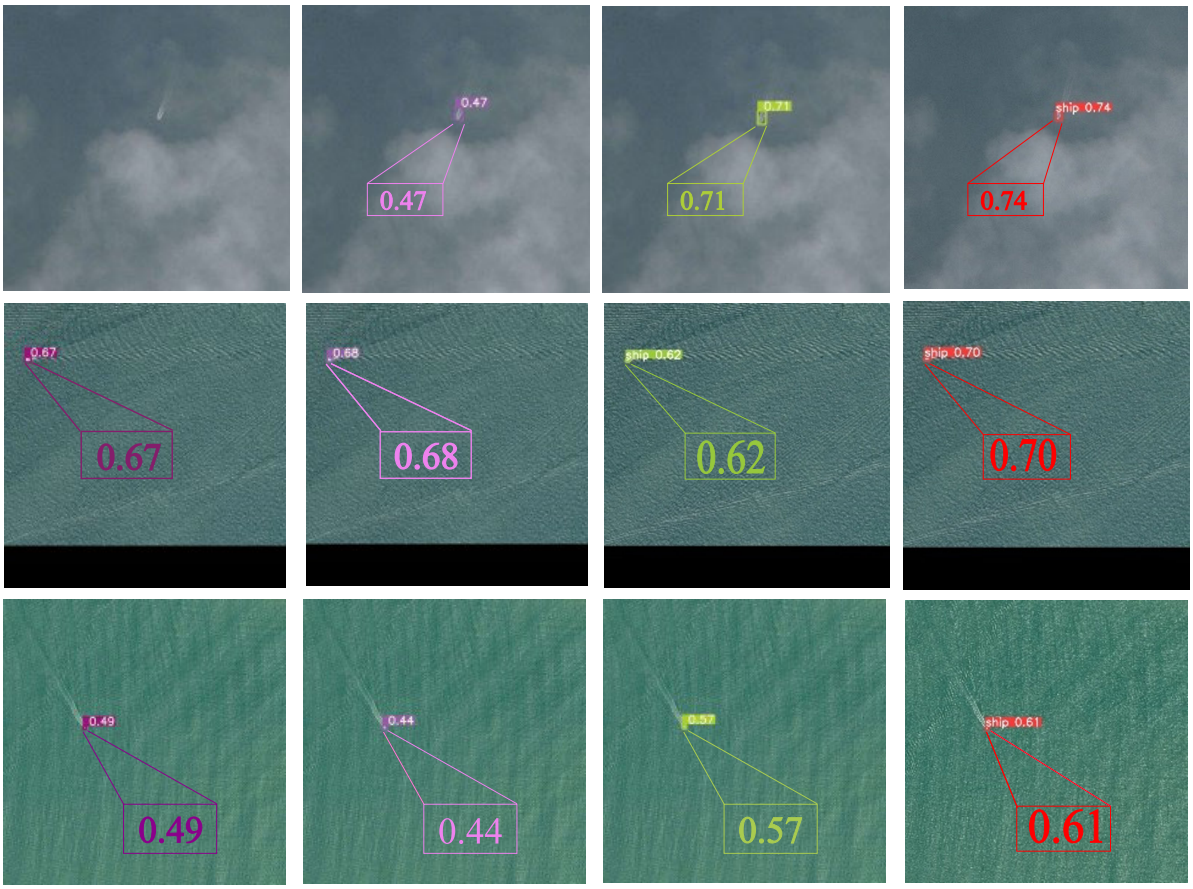


Fig. 11. Visualization comparison between our proposed method and three SOTA methods on the Airbus-Ship dataset.

posed MESA module, whose unique dual-domain attention mechanism enables the model to simultaneously focus on fine-grained features of small targets through spatial attention while capturing the broader contextual information required for larger targets via multiscale attention. Meanwhile, as shown in Fig. 11, in scenarios with complex backgrounds, our method successfully detected a target that was missed by YOLOv5s, achieving a higher confidence score compared to these comparative methods. In addition, as shown in Fig. 12, various comparative models exhibit missed detections and false alarms in scenarios with multiple small targets coexisting. In contrast, the method proposed herein achieved comprehensive identification and precise localization of all targets. This advantage is primarily attributed to our AFEM module, which is specifically designed to enhance the representation of small targets. By employing its heterogeneous multibranch structure,

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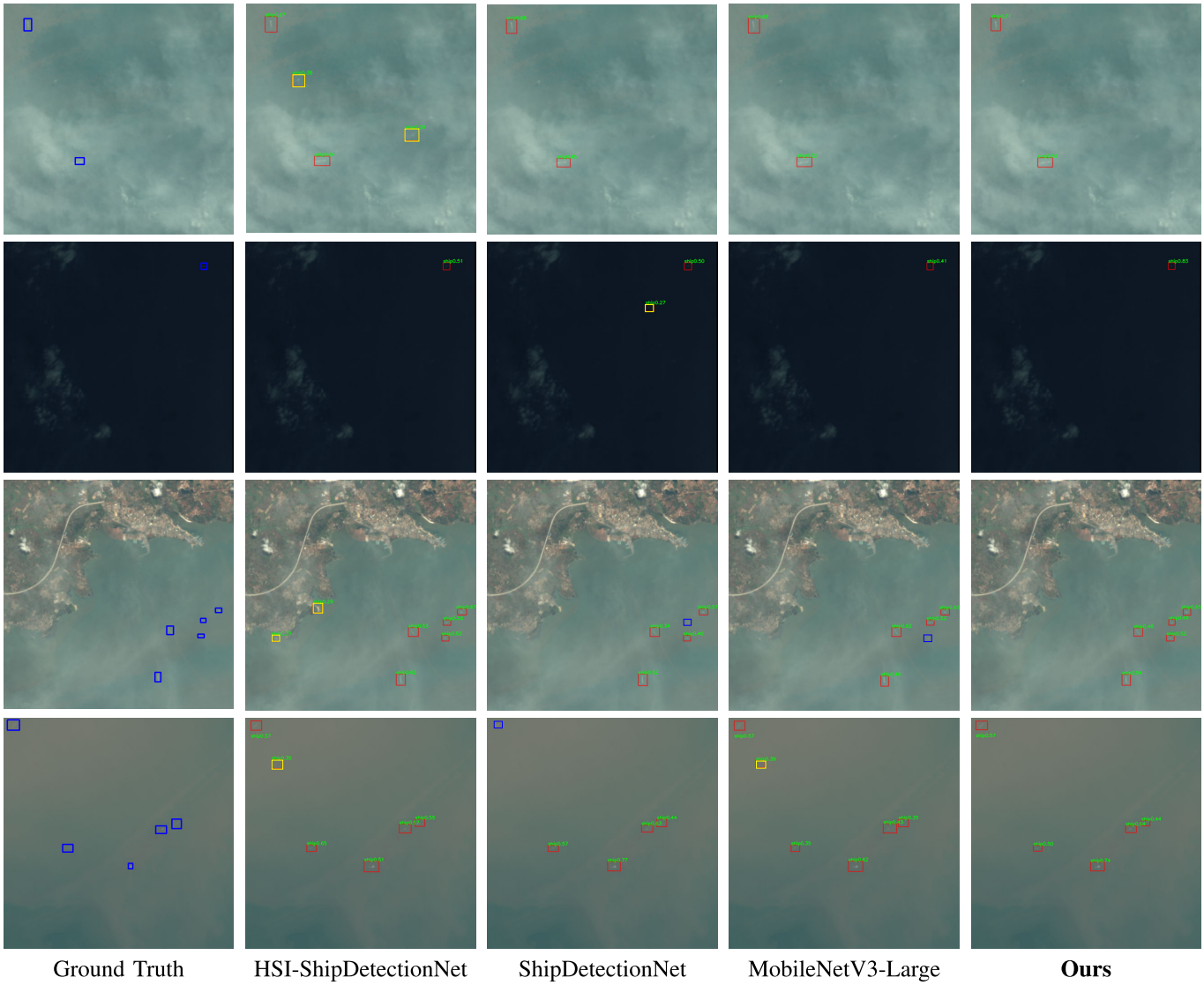


Fig. 12. Visualization comparison between our proposed method and three SOTA methods on the LEVIR-Ship dataset. The yellow annotations represent false positives, while the blue annotations indicate missed targets.

AFEM effectively preserves and strengthens the faint, fine-grained features of small objects in the shallow layers, thereby significantly improving the model's ability to distinguish and accurately localize them in complex scenes.

In summary, the proposed method achieves an optimal tradeoff between detection performance and computational cost, while demonstrating superior adaptability for small-scale ship detection tasks in multiscale environments.

D. Performance Evaluation in Complex Scenarios

To evaluate the operational reliability of the proposed model in real-world maritime environments, we conducted qualitative comparative experiments across multiple complex scenarios, as shown in Fig. 13.

First, the model's robustness under meteorological interference is analyzed. In the first row of Fig. 13, the target ship is partially veiled by thin cloud cover, resulting in severe spectral attenuation and contrast degradation. Despite these low-visibility conditions, LSDN still achieves high-confidence localization, validating that the AFEM module effectively captures weak textural cues even under atmospheric occlusion. In contrast, comparison models exhibit noticeably lower confidence scores. Second, the detection performance under thick cloud interference is examined. The second row of Fig. 13 depicts a more extreme condition where high-reflectance thick cumulus clouds create intense background clutter. In this scenario, the bright cloud edges often act as "pseudo-targets," leading to multiple false alarms in models such as YOLOv13s. Our proposed model demonstrates superior clutter suppression capability, accurately localizing the true ship with a reliable score of 0.53 without being misled by nontarget cloud masses. Third, we investigated the impact of intense sun glint and specular reflections. As illustrated in the third row of Fig. 13, the numerous bright reflection points on the sea surface create chaotic background noise. While HSI-ShipDetectionNet exhibits a false alarm by misidentifying a wave reflection as a ship, and YOLOv13s yields a lower confidence score, our model achieves a superior confidence score of 0.84.

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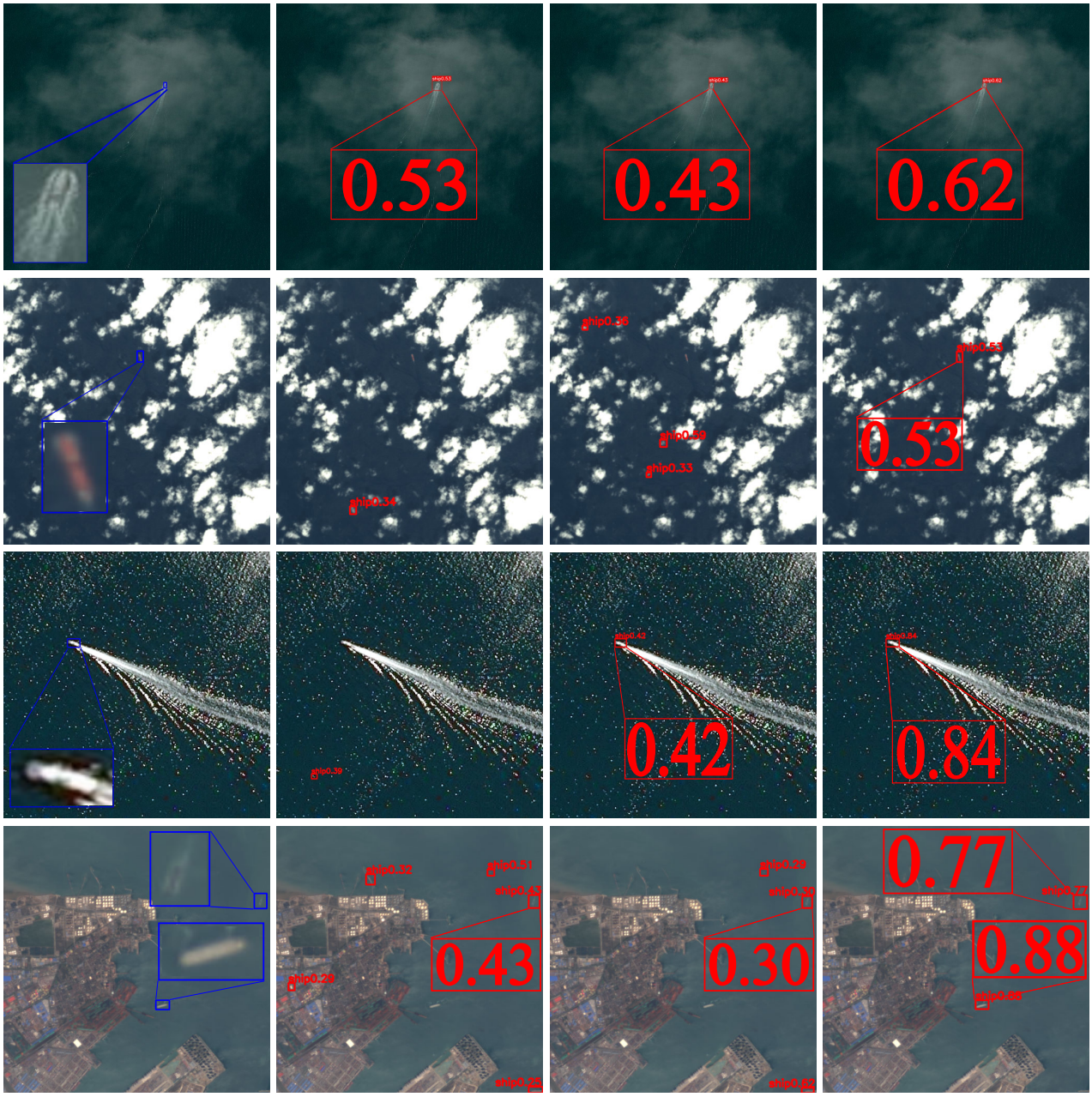


Fig. 13. Visualization comparison with different models in complex scenarios.

TABLE II
COMPARISON OF MODEL COMPLEXITY AND INFERENCE SPEED ON THE AIRBUS-SHIP DATASET

Methods	Para(M) ↓	GFLOPs ↓	Latency(ms) ↓	FPS ↑	mAP(%)
YOLOv3-tiny	8.67	12.9	5.75	120	73.28
YOLOv5s	7.05	16.3	9.38	87	74.25
YOLOv12s	9.30	19.6	6.38	93	76.53
YOLOv11s	9.43	21.5	**4.38**	**157**	76.01
YOLOv13s	9.02	21.0	8.45	107	76.70
Ours	**3.65**	**2.5**	10.78	62	**81.60**

This demonstrates that LSDN effectively distinguishes the rigid structural features of ships from high-intensity solar-induced clutter. Finally, the model's discriminative capability is verified in complex littoral scenarios. The fourth row of Fig. 13 presents results in a scene containing dense terrestrial infrastructures, which present significant challenges due to their morphological similarity to ships. While comparison models suffer from significant confidence degradation or false

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TABLE III
ABLATION STUDY OF THE PROPOSED METHOD ON THE AIRBUS-SHIP DATASET

MESA	Dynamic Anchor Point	AFEM	Para(M) ↓	GFLOPs ↓	Size(MB) ↓	mAP(%)
×	×	×	**3.52**	**2.0**	**7.6**	72.80
✓	×	×	3.55	2.1	7.8	78.90
×	✓	×	3.52	2.0	7.6	76.80
×	×	✓	3.63	2.3	7.9	78.50
✓	✓	×	3.55	2.1	7.8	78.93
✓	×	✓	3.65	2.5	8.0	81.10
×	✓	✓	3.63	2.3	7.9	80.80
✓	✓	✓	3.65	2.5	8.0	**81.60**

positives on land, our method precisely identifies ship targets with high confidence while maintaining a zero false-alarm rate on land-based structures. This superior performance confirms the robustness and reliability of our model in diverse and high-clutter real-world backgrounds.

E. Analysis of Complexity and Inference Efficiency

To analyze the discrepancy between theoretical complexity and practical inference speed, we benchmarked several representative models from the YOLO series. The results, as shown in Table II, reveal that a model's real-world performance is critically influenced by its architectural design, rather than by theoretical indicators such as GFLOPs or parameter count alone.

For instance, YOLOv3-tiny, despite having a larger number of parameters, outperforms YOLOv5s in terms of FPS. This phenomenon can be attributed to its simple and hardware-friendly backbone composed primarily of standard convolution and pooling operations, which enables more efficient GPU utilization. In contrast, the complex feature integration mechanisms used in YOLOv5s, including cross-stage connections and multipath feature aggregation, introduce additional computational and memory-access overhead, which limits its practical inference speed. This principle continues to evolve in SOTA architectures such as YOLOv11s, which establishes a new performance benchmark by combining hardware efficiency with algorithmic optimization. It adopts a hardware-aware backbone that improves execution efficiency and an anchor-free detection head that reduces postprocessing bottlenecks such as nonmaximum suppression. These improvements demonstrate that achieving peak performance requires a careful balance between architectural design and computational efficiency.

Our method achieves the highest mAP of 81.6% on the Airbus-Ship dataset with only 3.65-M parameters and 2.5 GFLOPs. In addition, its performance regarding latency and FPS is comparable to other models. This performance profile reflects a deliberate design tradeoff we have adopted: prioritizing model lightness and ease of deployment over maximizing raw speed on high-end GPUs. In resource-constrained edge device scenarios, where power, memory, and computational units are strictly limited, the model's parameter size and computational load (GFLOPs) are typically the core factors determining project feasibility. Therefore, our design philosophy focuses on achieving dual efficiency in both theory and deployment under constrained hardware budgets, rather than pursuing an absolute speed advantage on powerful GPUs. This approach is perfectly consistent with the core objective stated in our article: to design lightweight detectors for resource-constrained platforms.

TABLE IV
ABLATION OF THE MESA

Methods	Para(M) ↓	GFLOPs ↓	Size(MB) ↓	mAP(%)
LSDN (w/o MESA)	3.52	2.0	7.6	72.80
LSDN (w/ MESA)	3.55	2.1	7.8	**78.90**

F. Ablation Experiments

To ascertain the efficacy of our method, we conducted ablation studies on the three modules using the Airbus-Ship dataset, as shown in Table III. Below, we will analyze the ablation studies for each module.

1) Ablation of the MESA: To assess the utility of the MESA module in enhancing model capabilities, our preliminary tests involved using a detection network with MESA-ShipDetection as the backbone network, excluding the MESA module, and obtaining the results. Subsequently, the MESA module was integrated into MESA-ShipDetection. The experimental results, as shown in Table IV, indicate that the MESA module increased the model's Para, GFLOPs, and model size by only 0.85%, 5%, and 2.63%, respectively, demonstrating the lightweight nature of the module. In addition, the MESA module improved mAP by 8.38%. The experiment demonstrates that the MESA module dynamically adjusts feature weights through a multiscale and spatial dual-domain attention mechanism, enhancing the model's focus capability on multiscale targets, thereby improving detection accuracy.

To systematically validate the effectiveness of our proposed attention modules, we conducted separate ablation studies for the two core components of MESA, the MAB and the SAB, against their respective baseline modules. The results are shown in Fig. 14, where Para represents the additional parameters introduced relative to the baseline model.

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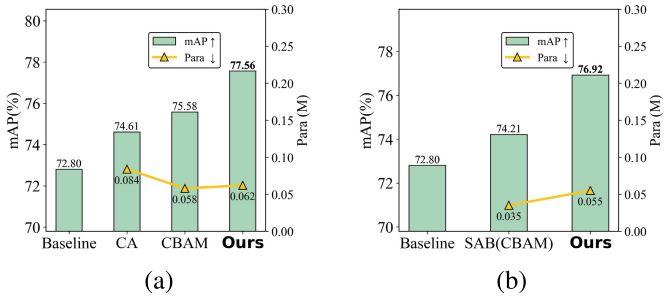


Fig. 14. Performance comparison of the two core components of MESA against their baseline modules. The Para in the figure represents the additional parameters introduced relative to the baseline model. (a) Performance comparison of our proposed MAB against the CA and CBAM modules on the baseline model. (b) Performance comparison of our proposed SAB against the original spatial attention module from CBAM on the baseline model.

First, as shown in Fig. 14(a), we compared the performance of our proposed MAB with the classic CA and CBAM modules. The experimental results clearly indicate that our MAB achieves a significant lead in detection accuracy, reaching an mAP of 77.56%, which is 2.95 and 1.98 percentage points higher than CA and CBAM, respectively. What is more noteworthy is that our MAB introduces only 0.062-M additional parameters, a count not only far lower than that of CA but also nearly on par with CBAM. In summary, our MAB increases the mAP by 6.54% relative to the baseline model while maintaining a lightweight profile. This robustly demonstrates that through our design of a parallel dual-branch structure and adaptive weighted fusion, we have achieved the optimal tradeoff between performance and efficiency.

Second, as shown in Fig. 14(b), our SAB achieves an mAP of 76.92%, which is much higher than the 74.21% achieved by the CBAM-based SAB and represents a 5.66% increase compared with the baseline model. This improvement strongly demonstrates the effectiveness of our structural modifications. Unlike CBAM, which prematurely compresses C channels of information into a single-channel representation, our SAB preserves the complete channel dimensionality and introduces independent convolutional paths for the dual-pooling branches, enabling it to generate a dedicated spatial attention map for each channel. This more refined and expressive attention mechanism can learn far more complex feature interactions than traditional shared attention approaches, ultimately leading to significant improvements in detection performance. Although the introduced parameter count of 0.055 M is slightly higher than SAB (CBAM)'s 0.035 M, the performance gains substantially outweigh the minimal cost increase.

2) Ablation of the Dynamic Anchor Points: We propose a dynamic anchor generation strategy, which constructs three sets of adaptive anchor parameters through the K-means clustering algorithm. To evaluate the impact of this dynamic anchor strategy on the model's detection performance, we conducted comparative experiments under the same dataset and training conditions against the fixed anchor parameters of the original YOLOv5. As shown in Table V, our proposed dynamic anchor strategy improved the mAP by 4 percentage points compared to the original fixed anchors, demonstrating

TABLE V
ABLATION OF THE DYNAMIC ANCHOR POINTS

Methods	mAP(%)
LSDN (Original Anchor Point)	72.80
LSDN (Dynamic Anchor Point)	**76.80**

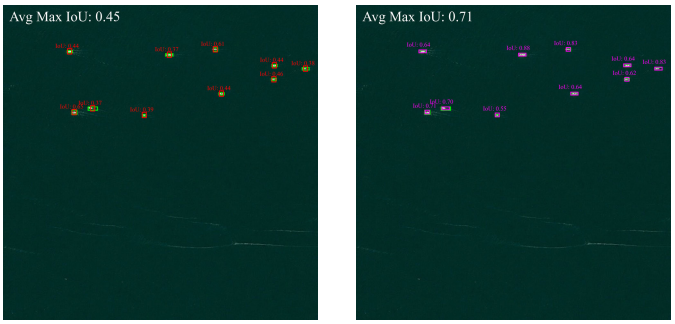


Fig. 15. Visualization comparison of anchor matching quality in a multitarget test scene. (a) Best-fit anchors from the original fixed anchor set. (b) Best-fit anchors from the dynamically generated anchor set.

TABLE VI
ABLATION OF THE AFEM AT DIFFERENT POSITIONS

	Para(M) ↓	GFLOPs ↓	Size(MB) ↓	mAP(%)
Stage=1	3.63	2.3	7.9	**78.50**
Stage=2	3.63	2.3	7.9	76.40
Stage=3	3.93	2.3	8.5	76.90
Stage=4	3.93	2.3	8.5	77.80
Stage=5	5.12	2.3	10.9	77.80
w/o AFEM	**3.52**	**2.0**	**7.6**	72.80

that dynamic anchors can better adapt to the detection requirements of multiscale targets. By leveraging clustering analysis based on the actual sizes of targets in the dataset, dynamic anchors can more flexibly match targets of varying scales and aspect ratios.

To fundamentally explain and visually demonstrate the reason for this performance gain, we conducted a further visual comparative analysis of the anchor matching quality, as shown in Fig. 15. This figure displays the best possible matches of the original anchors and our dynamic anchors with the GT Boxes in the same typical multitarget scene. As shown in Fig. 15(a), the matching quality between the original anchors and the ground truth targets is generally low, with an Average Max IoU (Avg Max IoU) of only 0.45 in this scene. In contrast, as shown in Fig. 15(b), our dynamically generated anchors are able to achieve a high degree of overlap with the real targets, boosting the Average Max IoU significantly to 0.71. This visual evidence clearly indicates that the superior matching quality established at the beginning of the training process directly provides better initial position estimates for the subsequent detection task, which ultimately translates into a significant improvement in accuracy.

3) Ablation of the AFEM: As shown in Table VI, we inserted the AFEM module into the various stages comprising the backbone (from Stage 1 to Stage 5) and recorded the

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TABLE VII
ABLATION OF AFEM WITH OTHER MODULES

Methods	Para(M) ↓	GFLOPs ↓	Size(MB) ↓	mAP(%)
w/o AFEM	**3.52**	**2.0**	**7.6**	72.80
w/ RepVGG [51]	3.67	2.5	7.9	76.04
w/ SPPCSPC [52]	3.94	3.4	8.5	77.58
w/ RFB-s [53]	3.56	2.1	7.7	75.72
w/ AFEM	3.63	2.3	7.9	**78.50**

experimental results. To independently study the impact of the AFEM module, the experimental network excluded the MESA module and the dynamic anchor strategy. We compared the results of each AFEM module insertion with the initial network (without the AFEM module). The experimental results indicate that the increase in GFLOPs was consistently 15%. When the AFEM module was inserted at Stage 5, the model's Para and size increased the most, by 45.45% and 43.42%, respectively. Similarly, when the module was inserted at Stage 3 and Stage 4, the increases in Para and size remained relatively high, at 11.65% and 11.84%, respectively. On the other hand, although the increases in Para and size were the same when the module was inserted at Stage 1 and Stage 2, at 3.13% and 3.95%, respectively, the mAP improvement was the highest at 7.83% when inserted at Stage 1. This experiment demonstrates that the AFEM module enhances shallow feature extraction capabilities through a multilevel feature fusion mechanism, effectively improving the feature information for small targets, thereby markedly boosting its detection accuracy. To further highlight the superiority of AFEM in enhancing features for small objects, we conducted a comparative ablation study against three other modules that also employ multilevel feature fusion. As shown in Table VII, the experimental results indicate that while all compared modules improve performance over the baseline model, AFEM achieves the most significant gains. Specifically, AFEM increases the mAP by 7.83% with only a 3.13% increase in parameters. This exceptional performance advantage is attributed to the unique designs of AFEM at both the feature extraction and feature fusion levels.

First, at the feature extraction level, the RepVGG Block enhances feature representation through parallel standard 3×3 and 1×1 convolutions, which is a homogeneous multibranch structure. However, for small targets whose features are inherently sparse, it is difficult to extract sufficient information by solely relying on standard square-shaped kernels. To address this, AFEM introduces asymmetric convolution (1×3 and 3×1) branches, enabling it to more finely capture the local edge and texture details of small targets. At the same time, AFEM also integrates dilated convolution branches. Without increasing computational cost, this significantly expands the receptive field, thereby enabling the capture of broader contextual information, which is crucial for distinguishing tiny ships from the background in complex sea conditions. Second, at the feature fusion level, both SPPCSPC and RFB-s employ static channel concatenation to integrate the outputs of their multiscale branches, which cannot differentiate based on the input content. However, for different scenarios, the importance of different feature types varies dynamically. To address this, AFEM adopts an adaptive weighted feature fusion mechanism. This allows the model to autonomously strike a balance between focusing on details and understanding the global context, preferentially amplifying the most useful features for the current scene while simultaneously suppressing interference from noise features.

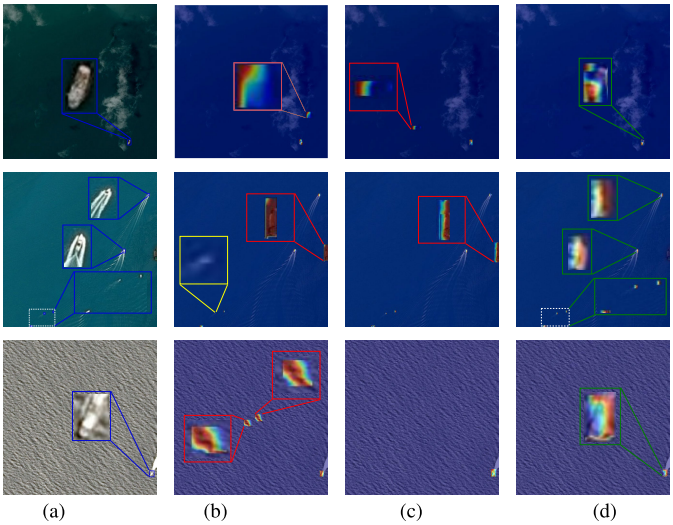


Fig. 16. Visualization comparison of feature activation maps from different enhancement modules. The red annotations represent false positives, while the yellow annotations indicate missed targets. (a) Ground.truth (b) RepVGG block. (c) SPPCSPC. (d) Ours.

As shown in Fig. 16, when processing complex scenarios such as those with cloud occlusion, multiple targets, and strong texture noise, the compared RepVGG Block and SPPCSPC commonly exhibit issues like diffuse activation regions, missed detections, or false alarms. In contrast, our proposed AFEM demonstrates superior robustness and discriminative power across all scenarios. This is attributed to its heterogeneous branches, which are capable of capturing diverse target features to avoid missed detections, while its adaptive fusion mechanism effectively suppresses background interference to reduce false alarms. Consequently, AFEM generates more concentrated and discriminative activation regions, enabling it to clearly separate tiny targets from complex environments. This series of visual results further validates that the structural design of AFEM possesses stronger robustness in remote sensing imagery compared to existing modules, which fundamentally explains its ultimate performance advantage.

V. CONCLUSION

In this article, we propose an innovative lightweight network, LSDN, designed for detecting ships in optical remote sensing imagery. We have constructed a lightweight detection model, MESA-ShipDetection, to replace the traditional CNN backbone network, and introduced a MESA, which improves its attention toward multiscale targets. In addition, we designed an AFEM module that employs a multilevel feature fusion mechanism to strengthen shallow feature extraction capabilities, thereby effectively enhancing the feature representation of

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small targets. Finally, we proposed a dynamic anchor generation strategy to achieve dynamic optimization and matching of anchor box sizes and ratios. Comparative experiments with other SOTA methods on two public ship datasets demonstrate that our method exhibits outstanding performance in detecting small and multiscale ships, with significant advantages in terms of lightweight design. In the future, we will incorporate more efficient feature extraction networks and cross-domain knowledge transfer techniques to improve the model's adaptability to different remote sensing data sources.

REFERENCES

[1] X. Li et al., "TKP-Net: A three keypoint detection network for ships using SAR imagery," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 364–376, 2024.

[2] K. Eldhuset, "An automatic ship and ship wake detection system for spaceborne SAR images in coastal regions," *IEEE Trans. Geosci. Remote Sens.*, vol. 34, no. 4, pp. 1010–1019, Jul. 1996.

[3] Y. Yin, X. Cheng, F. Shi, M. Zhao, G. Li, and S. Chen, "An enhanced lightweight convolutional neural network for ship detection in maritime surveillance system," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 5811–5825, 2022.

[4] B. Zhang, J. Liu, R. W. Liu, and Y. Huang, "Deep-learning-empowered visual ship detection and tracking: Literature review and future direction," *Eng. Appl. Artif. Intell.*, vol. 141, Feb. 2025, Art. no. 109754.

[5] B. Lebona, W. Kleynhans, T. Celik, and L. Mdakane, "Ship detection using VIIRS sensor specific data," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2016, pp. 1245–1247.

[6] X. Fan, Z. Hu, Y. Zhao, J. Chen, T. Wei, and Z. Huang, "A small-ship object detection method for satellite remote sensing data," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 11886–11898, 2024.

[7] G. Cheng, X. Xie, J. Han, L. Guo, and G.-S. Xia, "Remote sensing image scene classification meets deep learning: Challenges, methods, benchmarks, and opportunities," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, pp. 3735–3756, 2020.

[8] K. Ouchi, "Current status on vessel detection and classification by synthetic aperture radar for maritime security and safety," in *Proc. Int. Symp. Remote Sens. Environ. Sci.*, Feb. 2016, pp. 3–5.

[9] Z. Sun et al., "Arbitrary-direction SAR ship detection method for multiscale imbalance," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5208921.

[10] M. Yasir et al., "Ship detection based on deep learning using SAR imagery: A systematic literature review," *Soft Comput.*, vol. 27, no. 1, pp. 63–84, Jan. 2023.

[11] X. Leng, K. Ji, and G. Kuang, "Ship detection from raw SAR echo data," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5207811.

[12] K. Simonyan and A. Zisserman, "Very deep convolutional networks for large-scale image recognition," 2014, *arXiv:1409.1556*.

[13] S. Ren, K. He, R. Girshick, and J. Sun, "Faster R-CNN: Towards real-time object detection with region proposal networks," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 39, no. 6, pp. 1137–1149, Jun. 2017.

[14] J. Redmon, S. Divvala, R. Girshick, and A. Farhadi, "You only look once: Unified, real-time object detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 779–788.

[15] T. Y. Lin, P. Dollàr, R. Girshick, K. He, B. Hariharan, and S. Belongie, "Feature pyramid networks for object detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jul. 2017, pp. 2117–2125.

[16] W. Liu et al., "SSD: Single shot multibox detector," in *Proc. 14th Eur. Conf. Comput. Vis.*, Oct. 2016, pp. 21–37.

[17] C. Qin, X. Wang, Y. Liu, and G. Li, "A novel end-to-end transformer network for small scale ship detection in SAR images," in *Proc. IGARSS - IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2024, pp. 8158–8162.

[18] T. Chen et al., "MiM-ISTD: Mamba-in-mamba for efficient infrared small target detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5007613.

[19] N. Ma, L. Sun, Y. He, C. Zhou, and C. Dong, "CNN-TransNet: A hybrid CNN-transformer network with differential feature enhancement for cloud detection," *IEEE Geosci. Remote Sens. Lett.*, vol. 20, pp. 1–5, 2023.

[20] A. Dosovitskiy et al., "An image is worth 16×16 words: Transformers for image recognition at scale," 2020, *arXiv:2010.11929*.

[21] Y. He, B. Tu, P. Jiang, B. Liu, J. Li, and A. Plaza, "IGroupSS-mamba: Interval group spatial–spectral mamba for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5538817.

[22] H. Dong, F. Wei, Q. Xu, X. Wang, and G. Sun, "An end-to-end network for multi-scale ship detection in SAR images," in *Proc. 2nd Int. Symp. Control Eng. Robot. (ISCER)*, Feb. 2023, pp. 98–101.

[23] S. Li, X. Chen, and P. Lin, "A lightweight CNN and spatial-channel transformer hybrid network for image super-resolution," in *Proc. IEEE Int. Conf. Multimedia Expo (ICME)*, Jul. 2024, pp. 1–6.

[24] H. Song, L. Wang, L. Mi, H. Yu, and J. Gan, "DCAPT-Net: A multi-scale ship detection algorithm for SAR images," in *Proc. 10th Int. Conf. Big Data Inf. Analytics (BigDIA)*, Oct. 2024, pp. 441–446.

[25] X. Yang, J. Zhang, C. Chen, and D. Yang, "An efficient and lightweight CNN model with soft quantification for ship detection in SAR images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5230713.

[26] Z. Li, Y. You, and F. Liu, "Analysis on saliency estimation methods in high-resolution optical remote sensing imagery for multi-scale ship detection," *IEEE Access*, vol. 8, pp. 194485–194496, 2020.

[27] D. Liu, J. Zhang, Y. Qi, Y. Wu, and Y. Zhang, "Tiny object detection in remote sensing images based on object reconstruction and multiple receptive field adaptive feature enhancement," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5616213.

[28] Y. Yin, X. Cheng, F. Shi, X. Liu, H. Huo, and S. Chen, "High-order spatial interactions enhanced lightweight model for optical remote sensing image-based small ship detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 4201416.

[29] X. Zhang, X. Zhou, M. Lin, and J. Sun, "ShuffleNet: An extremely efficient convolutional neural network for mobile devices," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 6848–6856.

[30] N. Ma, X. Zhang, H.-T. Zheng, and J. Sun, "ShuffleNet V2: Practical guidelines for efficient CNN architecture design," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2018, pp. 116–131.

[31] F. Chollet, "Xception: Deep learning with depthwise separable convolutions," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 1251–1258.

[32] F. N. Iandola, S. Han, M. W. Moskewicz, K. Ashraf, W. J. Dally, and K. Keutzer, "SqueezeNet: AlexNet-level accuracy with 50× fewer parameters and <0.5MB model size," 2016, *arXiv:1602.07360*.

[33] A. G. Howard et al., "MobileNets: Efficient convolutional neural networks for mobile vision applications," 2017, *arXiv:1704.04861*.

[34] M. Sandler, A. Howard, M. Zhu, A. Zhmoginov, and L.-C. Chen, "MobileNetV2: Inverted residuals and linear bottlenecks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 4510–4520.

[35] A. Howard et al., "Searching for MobileNetV3," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 1314–1324.

[36] L. Wang et al., "A lightweight CNN for multi-source infrared ship detection from unmanned marine vehicles," *Heliyon*, vol. 10, no. 4, Feb. 2024, Art. no. e26229.

[37] J.-C. Zheng, S.-D. Sun, and S.-J. Zhao, "Fast ship detection based on lightweight YOLOv5 network," *IET Image Process.*, vol. 16, no. 6, pp. 1585–1593, May 2022.

[38] H. Changhua, C. Chen, and H. Chuan, "Ship small target detection based on deep convolution neural network in SAR image," *J. Chin. Inertial Technol.*, vol. 27, no. 3, pp. 397–405, 2019.

[39] L. Tian, Y. Cao, B. He, Y. Zhang, C. He, and D. Li, "Image enhancement driven by object characteristics and dense feature reuse network for ship target detection in remote sensing imagery," *Remote Sens.*, vol. 13, no. 7, p. 1327, Mar. 2021.

[40] A. Vaswani et al., "Attention is all you need," in *Proc. Adv. Neural Inf. Process. Syst.*, 2017, pp. 1–17.

[41] J. Chen, K. Chen, H. Chen, Z. Zou, and Z. Shi, "A degraded reconstruction enhancement-based method for tiny ship detection in remote sensing images with a new large-scale dataset," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5625014.

[42] C. Cui et al., "PP-LCNet: A lightweight CPU convolutional neural network," 2021, *arXiv:2109.15099*.

[43] K. Han, Y. Wang, Q. Tian, J. Guo, C. Xu, and C. Xu, "Ghostnet: More features from cheap operations," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2020, pp. 1580–1589.

[44] M. Tan and Q. Le, "EfficientNet: Rethinking model scaling for convolutional neural networks," in *Proc. Int. Conf. Mach. Learn.*, 2019, pp. 6105–6114.

[45] J. Redmon and A. Farhadi, "YOLOv3: An incremental improvement," 2018, *arXiv:1804.02767*.

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[46] Y. Tian, Q. Ye, and D. Doermann, "YOLOv12: Attention-centric real-time object detectors," 2025, *arXiv:2502.12524*.
[47] M. Lei et al., "YOLOv13: Real-time object detection with hypergraph-enhanced adaptive visual perception," 2025, *arXiv:2506.17733*.
[48] S. Ma, Y. Zhang, L. Peng, C. Sun, B. Ding, and Y. Zhu, "OWRT-DETR: A novel real-time transformer network for small-object detection in open-water search and rescue from UAV aerial imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 4205313.
[49] Z. Liu et al., "Improved YOLOv5s for small ship detection with optical remote sensing images," *IEEE Geosci. Remote Sens. Lett.*, vol. 20, pp. 1–5, 2023.
[50] T. Lin et al., "Microsoft COCO: Common objects in context," in *Proc. Eur. Conf. Comput. Vis.*, 2014, pp. 740–755.
[51] X. Ding, X. Zhang, N. Ma, J. Han, G. Ding, and J. Sun, "RepVGG: Making VGG-style ConvNets great again," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 13733–13742.
[52] C.-Y. Wang, A. Bochkovskiy, and H.-Y.-M. Liao, "YOLOv7: Trainable bag-of-freebies sets new state-of-the-art for real-time object detectors," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 7464–7475.
[53] S. Liu and D. Huang, "Receptive field block net for accurate and fast object detection," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, Sep. 2018, pp. 385–400.



Shounan Wang received the bachelor's degree from Cangzhou Jiaotong College, Cangzhou, China, in 2024. He is currently pursuing the master's degree with the Key Laboratory of Computer Vision and System, Ministry of Education, Tianjin University of Technology, Tianjin, China.

His research interests include object detection and model lightweighting.



Chen Jia (Member, IEEE) received the Ph.D. degree from Tianjin University of Technology, Tianjin, China, in 2023.

He is currently a Post-Doctoral Researcher with Tianjin University of Technology. He has published more than 15 articles as first and co-author in his research interests, including light field imaging and pattern recognition.



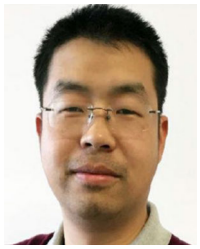
Hui Liu (Graduate Student Member, IEEE) received the bachelor's degree from the Qilu University of Technology (Shandong Academy of Sciences), Jinan, China, in 2023. He is currently pursuing the master's degree with the Key Laboratory of Computer Vision and System, Ministry of Education, Tianjin University of Technology, Tianjin, China.

His research interests include semantic segmentation and model lightweighting.



Yifan Yin (Student Member, IEEE) received the B.S. and M.S. degrees in computer science and technology from Tianjin University of Technology, Tianjin, China, in 2019 and 2023, respectively. She is currently pursuing the Ph.D. degree with the University of Technology Sydney, Sydney, NSW, Australia.

Her research interests include artificial intelligence, deep learning, computer vision, and object detection.



Xu Cheng (Senior Member, IEEE) received the Ph.D. degree in engineering from the Department of Ocean Operations and Civil Engineering, Intelligent Systems Laboratory, Norwegian University of Science and Technology, Ålesund, Norway, in June 2020.

From June 2020 to March 2022, he was a Post-Doctoral Fellow and a Researcher with the Department of Manufacturing and Civil Engineering, Gjøvik, Norway. From April 2022 to June 2023, he was a Permanent Researcher with Smart Innovation Norway, Halden, Norway. He is currently a Full Professor with Tianjin University of Technology, Tianjin, China. He has applied for and coordinated more than five projects supported by the Norwegian Research Council, the EU, and industry. He has published more than 60 articles as first and co-author in his research interests, including data analysis and artificial intelligence in maritime operations, time series analysis, and predictive maintenance of wind turbines.



Shengyong Chen (Senior Member, IEEE) received the Ph.D. degree in computer vision from the City University of Hong Kong, Hong Kong, in 2003.

From 2006 to 2007, he received a Fellowship from the Alexander von Humboldt Foundation of Germany and worked with the University of Hamburg, Hamburg, Germany. From 2008 to 2012, he worked as a Visiting Professor with Imperial College London, London, U.K., and the University of Cambridge, Cambridge, U.K. He is currently a Full Professor with Tianjin University of Technology, Tianjin, China, and the Director of the Engineering Research Center of Learning-Based Intelligent Systems, Ministry of Education. He has been conducting research on vision sensors for robotics for more than 20 years. He has published over 300 scientific papers in international journals and conferences, including 80 articles in IEEE TRANSACTIONS. He also published more than ten books in the past years and applied for more than 200 patents.

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